Arnav Sharma

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EDUCATION

University of California, Davis Expected Jun 2027
B.S. in Computer Science · Minor in Business Studies
Dean's List: Spring 2024 | College of Engineering | Graduate School of Management
Relevant Coursework: Machine Learning, Artificial Intelligence, Computer Vision, Operating Systems, Computer Architecture

EXPERIENCE

Fonabit Technologies Pvt. Ltd. Jun 2025 – Sep 2025
Software Developer & Data Research Intern
- Built machine learning models to analyze dataset and generate insights that informed data-driven business decisions.
- Developed Tableau dashboards that translated model outputs into clear visual reports, reducing manual reporting effort by 70%.
- Partnered with the Salesforce support team to streamline organizational workflows.

Pixabits Technologies Pvt. Ltd. Jun 2024 – Aug 2024
Software Developer Intern
- Built backend services and REST APIs in Python and Java, improving response time along with throughput by over 50%.
- Debugged and performance-tested features alongside cross-functional teams across releases.
- Implemented responsive UI from Figma designs and refined components using user analytics, increasing engagement.

PROJECTS

CardioSense 2026 · UC Davis
- Built a sub-$30 wearable ECG that continuously detects atrial fibrillation on-device (no cloud), returning a rhythm verdict in under 10 seconds — built end-to-end in 24 hours.
- Trained a 43K-parameter 1D CNN (PyTorch) on MIT-BIH clinical ECG datasets to classify 10-second windows as Normal/Arrhythmia, reaching ~91% accuracy (95% with a strict "Uncertain" safety gate) at ~10 ms inference and a 200 KB model.
- Built the signal-processing pipeline (bandpass/notch filtering, Pan-Tompkins R-peak detection, HRV feature extraction) and real-time SwiftUI patient and Streamlit clinician interfaces streaming over BLE.

InfraCopilot AI 2026 · UC Davis
- Built a full-stack predictive-maintenance platform for EV charging networks.
- Developed a cost-aware ML model (scikit-learn, SMOTE) achieving ~90% failure recall and $300K+ in simulated savings.
- Engineered a FastAPI backend and Next.js dashboard delivering real-time risk scoring, root-cause insights, and actionable maintenance recommendations.

Fake News Classifier 2025 · UC Davis
- Built an end-to-end NLP pipeline assessing news credibility using TF-IDF + Logistic Regression and a DistilBERT transformer, trained on ~44K real-world articles.
- Achieved ~99% test accuracy and designed probabilistic credibility scoring (beyond binary labels), emphasizing rigorous evaluation and ethical model outputs.

TECHNICAL SKILLS

Languages: Python, Java, C, C++, JavaScript, Assembly
Machine Learning & AI: PyTorch, scikit-learn, Deep Learning, CNNs, Transformers (DistilBERT), Computer Vision (OpenCV), NLP, TF-IDF, SMOTE
Systems & Architecture: Operating Systems, Concurrency & Multithreading, Memory Management, CPU Pipelining, Cache/Memory Hierarchy, Embedded (Arduino/BLE)
Data & Backend: NumPy, Pandas, FastAPI, Next.js, Streamlit, REST APIs, Signal Processing (DSP)
Tools & Platforms: Git, Linux/Unix, Jupyter, Tableau, Salesforce, Figma